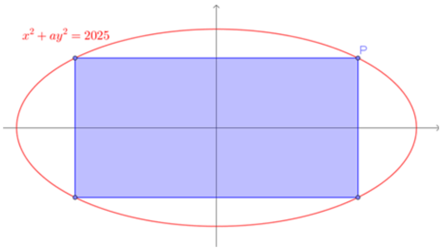
S1Q10

Question 10 - Stage 1



The point P lies on the ellipse with equation $x^2 + ay^2 = 2025$.

P is reflected in the x-axis to give point P', and both P and P' are reflected in the y-axis.

These four points are corners of a rectangle as shown.

If the maximum possible value of the area of a rectangle formed in this way is 2000 square units, what is the value of a (to 4 d.p.)?

Working

$$
\begin{aligned}
&P(45\cos(\theta), 45a^{-\frac{1}{2}}\sin(\theta)) \\
90\cos(\theta)\cdot 90a^{-\frac{1}{2}}\sin(\theta) &= 2000 \\
\cos(\theta)\cdot a^{-\frac{1}{2}}\sin(\theta) &= \frac{20}{81} \\
\cos(\theta)\sin(\theta) &= \frac{20}{81}a^{\frac{1}{2}} \\
\frac{1}{2}\sin(2\theta) &= \frac{20}{81}a^{\frac{1}{2}} \\
\sin(2\theta) &= \frac{40}{81}a^{\frac{1}{2}} \\
\text{max of } \sin(2\theta) &= 1 \\
\frac{40}{81}a^{\frac{1}{2}} &= 1 \\
a^{\frac{1}{2}} &= \frac{81}{40} \\
a &= \left(\frac{81}{40}\right)^2 \\
&\approx 4.1006
\end{aligned}
$$